floyd

人员

于家瑞、陶汇笙、王恩泽、郭栩睿、洪晨栋、洪晨棋、于霄龙 到课

上周作业检查

上周作业链接: https://cppoj.kids123code.com/contest/3584

王向东老师周六十点半C++拓扑排序

刷新

#	用户名	姓名	编程分	时间	A	B	C	D	E
1	hongchenqi	洪晨棋	400	693	100	100	100	100	
2	wangenze	王恩泽	400	711	100	100	100	100	
3	yujiarui	于家瑞	400	791	100	100	100	100	
4	hongchendong	洪晨栋	400	838	100	100	100	100	
5	taohuisheng	陶汇笙	372	582	100	100	100	72	
6	guoxurui	郭栩睿	336	582	100	100	100	36	
7	yuxiaolong	于霄龙	200	141	100	100			

本周作业

https://cppoj.kids123code.com/contest/3728 (课上讲了 A ~ D 这些题, 课后作业是 E 题, E 题得到暴力分即可, 不需要得满分)

课堂表现

今天讲的 floyd 属于图论里面比较简单的算法, 同学们上课整体都吸收、掌握的比较好。

课堂内容

口袋的天空 (上周作业)

按照最小生成树的方法来做, 直到剩 K 个联通块为止

```
#include <bits/stdc++.h>

using namespace std;

const int maxn = 1000 + 5;
struct node {
  int a, b, value;
  bool operator < (const node& p) const { return value < p.value; }
};
vector<node> edges;

int f[maxn];
```

```
int fFind(int x) {
  if (f[x] != x) f[x] = fFind(f[x]);
  return f[x];
}

int main()
{
  int n, m, K; cin >> n >> m >> K;
  while (m -- ) {
    int a, b, value; cin >> a >> b >> value; edges.push_back({a,b,value});
  }

  sort(edges.begin(), edges.end());
  for (int i = 1; i <= n; ++i) f[i] = i;

  if (n == K) { cout << 0 << endl; return 0; }

  int res = 0;
  for (node it : edges) {
    int a = it.a, b = it.b, value = it.value;
    int fa = fFind(a), fb = fFind(b);
    if (fa != fb) {
      n--; res += value; f[fa] = fb;
      if (n == K) { cout << res << endl; return 0; }
    }
  }

  cout << "No Answer" << endl;
  return 0;
}
```

【模板】Floyd

floyd 模板, 求多源最短路

f[i][j]: i 到 j 的最短路是多少

```
#include <bits/stdc++.h>

using namespace std;

const int maxn = 100 + 5;
const int inf = 0x3f3f3f3f;
int f[maxn][maxn];

int main()
{
  int n, m; cin >> n >> m;
  memset(f, 0x3f, sizeof(f));
  for (int i = 1; i <= n; ++i) f[i][i] = 0;
  while (m -- ) {
```

```
    int a, b, c; cin >> a >> b >> c;
    f[a][b] = f[b][a] = min(f[a][b], c);
  }

  for (int k = 1; k <= n; ++k) {
    for (int i = 1; i <= n; ++i) {
      for (int j = 1; j <= n; ++j) f[i][j] = min(f[i][j], f[i][k]+f[k][j]);
    }
  }

  for (int i = 1; i <= n; ++i) {
    for (int j = 1; j <= n; ++j) cout << f[i][j] << " ";
    cout << endl;
  }
  return 0;
}
```

[USACO07NOV] Cow Hurdles S

要求路径上最高的栏最小, 就在 floyd 的过程中, 把 dp 的方式改一下即可

改成 `f[i][j] = min(f[i][j], max(f[i][k],f[k][j]))` 即可

```
#include <bits/stdc++.h>

using namespace std;

const int maxn = 300 + 5;
const int inf = 0x3f3f3f3f;
int f[maxn][maxn];

int main()
{
  int n, m, T; cin >> n >> m >> T;

  memset(f, 0x3f, sizeof(f));
  for (int i = 1; i <= n; ++i) f[i][i] = 0;
  while (m -- ) {
    int a, b, c; cin >> a >> b >> c; f[a][b] = c;
  }

  for (int k = 1; k <= n; ++k) {
    for (int i = 1; i <= n; ++i) {
      for (int j = 1; j <= n; ++j) f[i][j] = min(f[i][j], max(f[i][k],f[k][j]));
    }
  }

  while (T -- ) {
    int a, b; cin >> a >> b;
    cout << (f[a][b]==inf ? -1 : f[a][b]) << endl;
  }
```

```
  return 0;
}
```

[USACO08JAN] Cow Contest S

f[i][j]: 代表 i 到 j 的大小是否已经知道了

跑 floyd, 然后明确所有点对的大小关系, 最后判断每个点跟其他点的关系是否都已知

```
#include <bits/stdc++.h>

using namespace std;

const int maxn = 100 + 5;
bool f[maxn][maxn];

int main()
{
  int n, m; cin >> n >> m;
  while (m -- ) {
    int a, b; cin >> a >> b; f[a][b] = true;
  }

  for (int k = 1; k <= n; ++k) {
    for (int i = 1; i <= n; ++i) {
      for (int j = 1; j <= n; ++j) {
        f[i][j] |= (f[i][k]&&f[k][j]);
      }
    }
  }

  int res = 0;
  for (int i = 1; i <= n; ++i) {
    int cnt = 0;
    for (int j = 1; j <= n; ++j) {
      if (j == i) continue;
      if (f[j][i] || f[i][j]) cnt++;
    }
    if (cnt == n-1) ++res;
  }
  cout << res << endl;
  return 0;
}
```

灾后重建

某个村庄能用的话, 说明可以用这个村庄作为中转站跑 floyd 了

从前往后依次看每个时间点, 把这个时间点之前的村庄加入中转站跑 floyd 即可

```cpp
#include <bits/stdc++.h>

using namespace std;

const int N = 200 + 5, M = 50000 + 5;
const int inf = 0x3f3f3f3f;
int w[N], f[N][N];

void update(int k, int n) {
  for (int i = 0; i < n; ++i) {
    for (int j = 0; j < n; ++j) f[i][j] = min(f[i][j], f[i][k]+f[k][j]);
  }
}

int main()
{
  int n, m; cin >> n >> m;
  for (int i = 0; i < n; ++i) cin >> w[i];

  memset(f, 0x3f, sizeof(f));
  for (int i = 0; i < n; ++i) f[i][i] = 0;
  while (m -- ) {
    int a, b, c; cin >> a >> b >> c;
    f[a][b] = f[b][a] = c;
  }

  int T, id = 0; cin >> T;
  while (T -- ) {
    int x, y, t; cin >> x >> y >> t;
    while (id<n && w[id]<=t) {
      update(id, n);
      ++id;
    }

    if (w[x]>t || w[y]>t || f[x][y]==inf) cout << -1 << endl;
    else cout << f[x][y] << endl;
  }
  return 0;
}
```